

The Math

The Construction of Differential Geometry from Topology and
Linear Algebra, through the Lens of Physical Descriptions

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Preface

Push a block across a floor, gently, and then less gently. For a while nothing moves at all: the force climbs and the block sits, absorbing the push without an inch of response. Then, at a threshold, it lets go all at once. The laws of this stubbornness are old — Amontons wrote them down in 1699, and Coulomb in 1785 separated the sticking from the sliding — and every reader has felt them through a heavy piece of furniture: the response of the block is not proportional to the push. It is nothing, and then it is everything.

Shine light on a metal and ask for electrons. Brighten a red lamp as much as you please — pour energy in — and nothing comes out, ever; switch to a dim violet one and the electrons leave at once. The observation is Lenard's, in 1902; the explanation is Einstein's, in 1905, and it carried the century: the metal does not respond to the amount of light accumulated but to whether each arriving quantum clears a threshold. Below the boundary, nothing; above it, release — and no patience with the red lamp substitutes for crossing.

Pull a bar of steel and plot the pull against the stretch. At first the plot is a clean straight line — the response proportional to the load, the regime Hooke announced in 1678 and Young made quantitative in 1807, the regime where the material meets force with obedience. Then the line reaches the yield point and stops being a line: the proportionality ends at a boundary, and past it the bar answers by a different law entirely. The formal language of that plot — stress against strain — is Cauchy's, from 1822, and its shape is the same two-act story: a linear zone, and an edge where the linear zone runs out.

Close a valve on a long pipe of moving water. Close it slowly and nothing of note happens. Close it fast and the pipe bangs like a struck bell — the water hammer, whose surge Joukowski measured and wrote into an equation in 1898. The moving column does not negotiate with the closed valve; its

answer arrives whole, as a wave, at the moment the boundary is imposed.

Four phenomena, four mechanisms, and no claim here that they share one physics — each has its own, cited above, and this book will not blur them. What they share is a *shape of response*, and the reader should hold that shape, because this series is organized by it: a zone where response is proportional and accumulation buys progress; a boundary where proportion ends; and, at the boundary, a response that is withheld and then arrives whole. Nothing, then everything. That shape — offered here as this preface's own framing image, nothing more — is the one the reader's pocket already demonstrates in time. A receiver listening to three satellites does not close in gradually on its position; it sticks, holding two candidate answers it cannot choose between, and it waits — time accumulating against the ambiguity like force against a stuck block, a temporal friction — until the fourth reading arrives, or a valid calibration stands in for it, and the fix releases, whole. Why exactly four, and why the sticking is structural rather than sloppy, is the arithmetic this preface now performs.

So before the construction begins, a number has to be earned, because this book is about to be organized by it.

How many soundings does it take to fix a point? The question is older than any instrument in this series; it is the surveyor's question, and its arithmetic is classical. One measured distance confines the unknown to a sphere around the station. A second distance cuts that sphere with another, and the intersection of two spheres is a circle. A third cuts again, and a circle meets a sphere in exactly two points. There the classical count stalls, and it stalls in an interesting way: three honest measurements, performed perfectly, leave not a haze of candidates but precisely two — a point and its mirror twin, utterly indistinguishable to the three stations that measured them. This is the binary result of trilateration, and it is structural, not sloppy: no repetition of the same three soundings, however careful, will ever split the pair, because the ambiguity lives in the geometry, not in the noise.

The solution is the fourth sounding. A fourth station, placed off the plane of the first three, is near one twin and far from the other; its single reading breaks the tie, and the point, at last, has been named. The arithmetic of the earning is worth saying aloud: three dimensions demand three ranges to come down to finitely many candidates, and the finitely many is exactly two — a binary digit of ambiguity — so one more measurement, the smallest possible act, decides it. Four is three plus the tiebreaker. The count is not a convention and not a preference; it is the smallest number of soundings

that turns ranging into naming, in a world of three dimensions. The reader carries the living example in a pocket: satellite navigation fixes a receiver from four satellites — the classical geometry just told, with the honest engineering footnote that the fourth reading is also made to pay for the receiver's imperfect clock, a second job that changes nothing about the first.

The number is now earned, and here is what it organizes. Across this series, four models are being developed — four descriptions ranged on the same unknown. Volume 1 opened its preface with a fence this book keeps: the electron, in these pages, is the name of a mathematical model, and the argument touches descriptions, never the particle in the world. It is exactly because the electron is a model that it can be sounded the way a surveyor sounds a point: each description is a station, and no single station names the thing. There is the geometric description, in which the model's content appears as curvature; the quantum description, in which it appears as states and their duals; the electron model itself, the device's own subject, the one description with a measured number to its name; and the fluid description — promised in the founding text of the apparatus before any of the others were built, and, at this writing, still the promise the series has yet to pay. Four stations. Three would leave the position ambiguous — a mirror pair of readings with no way to choose — and the whole series is, in this exact sense, a trilateration: it does not expect any single description to name its subject, and it does not stall at three.

This volume builds none of the four. It builds the ground they stand on — the common language of neighborhoods, charts, tensors, connections, and curvature in which such descriptions are written at all — and it builds that language entirely from cited mathematics, name and date attached, with nothing original anywhere in the construction. A surveyor does not invent trigonometry; a surveyor learns it, cites it, and points the instrument. That is this book's whole posture. The four soundings belong to the volumes around it; the mathematics belongs to the names in the back matter; and the reader, as in Volume 1, is owed this notice on the first page: what follows asks them to believe nothing new. It only asks them to watch the classical instruments assemble.

Chapter 1

The Local

Every statement a physicist makes is local. The reading on the bench is valid near the bench; the field is linear in a small enough region; the temperature is uniform across a small enough cell; the law holds in the neighborhood where it was tested and is extended beyond it only on further evidence. Nobody teaches this discipline as a separate subject, because it is not a doctrine — it is the shape of honest reporting. A measurement is an assertion about a vicinity, and the physicist already reasons in vicinities all day long. This chapter does one thing: it shows that the mathematics of vicinities was written down, once and for all, and that nothing in it asks the reader to believe anything they do not already practice.

The writing-down has a name and a date. In 1914, in the *Grundzüge der Mengenlehre*, Felix Hausdorff codified the idea of a space in terms of neighborhoods: to every point, a family of sets called its neighborhoods, subject to a short list of axioms — every neighborhood of a point contains the point; the intersection of two neighborhoods contains a neighborhood; any set containing a neighborhood is one; and inside every neighborhood of a point sits a neighborhood in which it is a neighborhood of each of its points. That last clause is the load-bearing one, and the reader should hear it in lab language: a region of validity contains, around each of its points, a smaller region of validity. Validity nests. That is the whole axiom, and the physicist has never worked a day without it.

The modern presentation turns the same content around and speaks of open sets — call a set open when it is a neighborhood of each of its points — and then the axioms compress to three: the union of any family of open sets is open, the intersection of finitely many is open, and the whole space and

the empty set are open. The compression is bookkeeping, not new content; the standard reference for the assembled subject is Kelley's *General Topology* of 1955, and an equivalent axiomatization through the closure operation was given by Kuratowski in 1922. One vocabulary note earns its keep here: a *topology* on a set is nothing but the chosen family of open sets — the declared inventory of which regions count as regions of validity — and a *topological space* is a set wearing such an inventory. No distance has been mentioned. That omission is not a gap; it is the subject.

Two workhorse notions now come for free, and both are exactly what their lab readings suggest.

Continuity first. A map between topological spaces is continuous when the preimage of every open set is open — when every demand for output-validity can be met by a region of input-validity. Read physically: a process is continuous when, to guarantee the result within a stated tolerance region, it suffices to control the input within some region. No rate, no bound, no modulus — those belong to metric refinements the physicist may add later. The bare notion is preservation of vicinity, and Hausdorff's framework states it without a single number.

Limits second. A sequence converges to a point when every neighborhood of that point, however small, contains all but finitely many terms of the sequence — for every region of validity around the target, the sequence eventually enters and stays. (The idea of framing limits abstractly predates even the neighborhood axioms: Fréchet's 1906 thesis built convergence into abstract spaces by other means, and Hausdorff's neighborhoods absorbed the lesson.) The reader should pause on what is absent: no norm measures how far a term is from the target, and none is needed. Approach is a topological fact — membership in ever-smaller declared regions — before it is ever a numerical one. Every ball-and-epsilon limit the physicist has computed was secretly this statement, with the balls supplied by a metric; the statement itself never needed the ruler, only the regions.

There is one more notion the vicinity language supports, and it is the one a measuring instrument forces: separation. Given two points of a space, does some open set contain one and not the other? If yes for every pair — every two distinct points are told apart by some region — the space satisfies what the trade calls the T0 condition; stronger separation grades follow (each point closed, T1; disjoint neighborhoods around distinct points, T2, the condition Hausdorff himself built into his axioms), and the taxonomy is laid out in the classical literature descending from Alexandroff and Hopf's 1935 *Topologie*.

The physical reading is immediate: the open sets are the space's stock of distinguishing observations, and separation asks whether the observations can tell the points apart at all.

And when they cannot, mathematics does not avert its eyes; it quotients. Points that no open set separates are declared one point; the space collapses along indistinguishability, and the result — the same observations, no phantom distinctions — satisfies T0 by construction. The construction is standard, it descends from the same classical literature, and it carries Kolmogorov's name: the T0 identification, in which a space is replaced by the space of what its own topology can tell apart. Nothing original is being said here; this is textbook material, and the only reason to slow down over it is that the reader is about to see the same shape in an unexpected place.

An illustration, not an identity. The instrument this series measures — the apparatus documented in Volume 1 — performs, at one load-bearing joint, an identification of exactly this shape: states that its reading cannot tell apart are declared one state, and the flagship number is read off the identified space. That is the T0 identification's shape, exhibited in a working device; it is offered here as an illustration of the construction the reader has just met, and as nothing more. The device does not thereby do topology, and this book does not claim it does. Illustration, not identity.

Take stock of what is now in hand, because the inventory is the chapter's point. Nearness, without distance. Continuity, without a rate. Limits, without a norm. Distinguishability, and the discipline of collapsing what cannot be distinguished. Every one of these is cited, classical, and older than the reader's textbooks; every one of them formalizes a habit the working physicist already has; and not one of them required a ruler. The ruler — the metric, the inner product, the entire apparatus of lengths — is famously going to be the *last* thing this construction adds, chapters from now, and the reader should begin suspecting already how much geometry can stand without it.

What the vicinity language cannot yet do is hold coordinates. The physicist's next reflex after "valid here" is to lay down axes and measure off components, and that reflex — the chart, the observer's coordinates, and the discipline of changing between observers — is the next chapter's subject.

This chapter ends where it began, with the reader's own practice: they have always worked locally; now the word has axioms, a name, and a date.

Chapter 2

The Chart

An observer lays down coordinates. That is not a mathematical event; it is a bench event — an origin chosen where the apparatus happens to sit, axes aligned with the table’s edges, units taken from the instrument at hand. A second observer, in a second lab, lays down different ones. Neither set is right, neither is wrong, and everything either of them publishes must survive translation into the other’s numbers, or it was an artifact of the furniture and not a fact about the world. The physicist knows this discipline in the bones: physics is what survives the change of observer. This chapter is that discipline, written down as mathematics — and, as in the last chapter, the writing-down asks the reader to believe nothing they do not already practice.

Begin with what coordinates actually are. A chart is a map — in the cartographer’s sense before the mathematician’s — from a region of the space under study onto a patch of ordinary number-space, one point to one address, addresses varying continuously in the sense the last chapter earned. The word “map” should be taken seriously: the chart is not the territory, it is a labeled picture of a region, and like any honest map it covers only its region. A space is called locally Euclidean when every point has a neighborhood — the last chapter’s word, doing its work — that some chart covers. Nothing global is promised. The sphere of the Earth admits no single flat map without tearing, and never needed one: an atlas of overlapping regional maps has always sufficed the navigator, and it suffices mathematics.

The load-bearing structure lives where two maps overlap. There, every point carries two addresses, and the rule converting one address into the other — the transition map — is a map from number-patch to number-patch: observer-to-observer translation, expressed entirely in coordinates. Now the

construction takes its one step, and the reader should watch how little it costs. Demand that the transitions be continuous, and the patchwork is a topological manifold. Demand more — that the transitions be differentiable, as maps of number-space, where differentiation already has its classical meaning — and the patchwork is a smooth manifold. Notice what the demand touches: not the space, the *translations*. Smoothness is imposed on the observer-to-observer dictionaries, which live in ordinary number-space where calculus has worked since Newton; the space itself then inherits a calculus through its charts, region by region, with the transition maps guaranteeing that what one observer computes another can re-derive. Differential geometry begins as a treaty among observers.

The idea has a birthday, and the story deserves telling at the grade this series tells its instrument-stories. In 1854 in Göttingen, Bernhard Riemann, twenty-seven and a candidate for the right to lecture, submitted — as custom required — three topics for his trial lecture, expecting, as custom also all but promised, that the faculty would take the first. His examiner was Gauss, and Gauss, against custom, chose the third: the foundations of geometry. What Riemann produced under that surprise, in a few pressed weeks, was the lecture “On the Hypotheses Which Lie at the Foundations of Geometry” — delivered the tenth of June, 1854, to a faculty audience mostly unequipped to follow it, with famously almost no formulas, and by the account that has come down to us it moved Gauss, who was not a man reports describe as easily moved. Riemann’s proposal was the one this chapter has been building: space as an n -fold extended magnitude, charted region by region, its deeper structure — the metric relations, the lengths — not given by logic in advance but to be determined, as he said, by experience. The lecture was published only in 1868, two years after his death, by Dedekind; the physicists took the rest of a century to spend what it contained. The reader should mark the physical lens Riemann himself ground: the question of which geometry space wears, he insisted, belongs to measurement, not to axiom. This series agrees, and holds the thought for later volumes.

Between Riemann’s lecture and the modern textbook stands one more name. The patchwork definition — charts, atlases, smooth transitions — was made fully precise by Hassler Whitney in 1936, and Whitney proved in the same stroke the theorem that keeps the abstraction honest: every smooth n -dimensional manifold, however abstractly assembled from patches, can be embedded in an ordinary Euclidean space of dimension $2n + 1$ — realized as an honest surface sitting in a big enough number-space. The abstract view

costs nothing, then: whatever a patchwork manifold is, it is never anything more exotic than a smooth surface seen without reference to the room it sits in. The assembled modern treatment is standard in the literature — Lee’s *Introduction to Smooth Manifolds* is the reference this series will cite for the details it declines to repeat — and nothing in it goes beyond the two demands already stated: charts that cover, transitions that are smooth.

So what, in the end, *is* the manifold, if every chart is somebody’s furniture? It is the residue — the content that survives every admissible change of chart, the facts on which all observers’ translations agree. That is not a poetic gloss; it is the working definition the construction just earned, and it is the mathematical spelling of the discipline the chapter opened with: physics is what survives the change of observer, and a manifold is what survives the change of chart. The reader should also notice, one more time, what is still absent. A manifold, even a smooth one, has no lengths. No distances, no angles, no areas; nothing so far prefers one chart’s scale to another’s. The ruler is still chapters away, and the amount of structure standing without it continues to grow.

One thing a chart does supply, at every point it covers, is the possibility of asking a pointed question: standing at this point, at this instant, in which direction — and how fast? The physicist’s word for the answer is velocity, and velocity does not need the whole space to be defined; it needs only an instant’s worth of room around one point. What velocities are, what lives beside them, and how the two great families of components the physicist writes — the raised indices and the lowered ones — arise from one construction, is the next chapter’s subject, and it is where the second of this book’s two title instruments, linear algebra, enters the shop.

Chapter 3

The Direction

Stand at one point of the last chapter's manifold and ask the physicist's most pointed question: a body passes through here, this instant — which way, and how fast? The answer exists before any observer lays down coordinates. The thrown stone does not consult a chart to have a velocity; the velocity is a fact at the point, an arrow with a direction and a rate, and every chart that covers the point merely reports it in that chart's own components. This chapter is about the arrow, the read-outs that measure the arrow, and the discovery — the physicist has been using it in index notation for a century — that arrows and read-outs respond to a change of observer in two opposite ways, and must never be confused.

First the arrow, built honestly from what previous chapters own. A curve through the point — a smooth path passing through it at time zero — has, in any chart, a column of component-rates: differentiate the coordinates along the path, the classical calculus of number-space, which the treaty of Chapter 2 licenses region by region. Two curves through the point may have the same component-rates in one chart; the transition maps being smooth, they then have the same component-rates in every chart — agreement is chart-independent, and so it is a fact about the point, not the furniture. A velocity is exactly such an agreement-class of curves, and the velocities at a point form a vector space: they add, they scale, and the dimension is the chart's dimension. The trade calls it the tangent space at the point; the modern formalization is standard and lives in the same references as the last chapter's (Lee's, for this series' purposes). The idea that directed magnitudes in n dimensions deserve their own algebra — addition of arrows, scaling of arrows, without any appeal to lengths — is far older, and its author paid

for being early: Hermann Grassmann published the calculus of extension in 1844, a schoolmaster writing outside the universities, and the mathematics of his century mostly declined to read him. This series cites him the way the trade eventually did: as the one who had the vector space first.

Now the read-out. An arrow is only half of a measurement; the other half is the thing that turns an arrow into a number. Let a quantity be defined near the point — a temperature, a potential, any scalar field the bench provides — and ask: at what rate does a body with this velocity see that quantity change? The answer is a number for each velocity, and it depends on the velocity linearly: twice the speed, twice the rate; the sum of two motions, the sum of the rates. A linear machine from arrows to numbers is called a covector, and the covectors at a point form their own vector space, of the same dimension, called the dual. The physical reading should be said plainly: covectors are the measuring devices of this subject. An arrow is a state of motion; a covector is an instrument reading applied to motion; and the pairing of the two — instrument applied to state, number out — is the elementary act from which everything later in this book is assembled. The physicist has met the structure in another guise: Dirac's formulation of the quantum theory, in his 1930 *Principles*, pairs kets with bras — states with the linear functionals that read them — and the bracket that gives the formalism its name is exactly this pairing, the dual space working in the physicist's own house.

Here is the chapter's centerpiece, and it is a derivation, not a decree. Change observers: a second chart covers the point, and the treaty supplies the dictionary — the transition map — with its matrix of partial derivatives, the Jacobian of the change, evaluated at the point. Follow the arrow first. Its new components come from differentiating the new coordinates along the same curve, and the chain rule — the same chain rule the reader has always used — says: new components equal the Jacobian applied to old components. Now follow the read-out. Its job is to report the same physical rate no matter the observer, so its components must change in exactly the compensating way: by the inverse of the Jacobian, arranged so that when instrument meets state the Jacobians cancel and the number — the measured rate — comes out unchanged. Two laws, then, and nobody chose them: components of arrows transform one way, components of read-outs the opposite way, and both fall out of the chain rule applied to the observers' dictionary. The trade's names are contravariant for the arrow's law and covariant for the read-out's, and the physicist's raised and lowered indices are precisely the bookkeeping of which

law a given column of components obeys. The calculus built on exactly this distinction — systems of components classified by their transformation law, with rules for combining them so that measured numbers stay observer-invariant — was published by Ricci and Levi-Civita in 1901, the absolute differential calculus, and it is the machinery general relativity would later be written in. Nothing about it was invented for relativity; it was waiting.

One more honest note completes the chapter, and it continues this book's longest-running theme. The reader's working habit converts freely between raised and lowered indices — between arrows and read-outs — and nothing in this chapter licenses the conversion. As matters stand, the two are different species: an arrow is a state, a read-out is an instrument, their spaces are merely of equal dimension, and no natural dictionary matches them element to element. The conversion the physicist performs daily is a purchase, and the coin it is bought with is precisely the ruler this construction has not yet admitted: a metric. When the metric arrives, chapters from now, it will do exactly this work — pair each arrow with its own read-out — and the reader will see index-lowering for what it is: not notation, but geometry, at a price the book will pay in the open. Until then the species stay separate, and the separation is not pedantry; it is the reason the two transformation laws exist at all.

What comes next is multiplication. One arrow in, one number out, was the covector; the objects of physics — the stresses, the fields, the curvatures this series is walking toward — take several arrows and read-outs at once, in slots, and answer with a number, linearly in each slot. The machines with slots are the tensors, the subject of the next chapter, and their first and oldest example was not invented by a geometer at all: it was found inside a loaded body, by Cauchy, holding the stresses of continuum matter — which is where the fourth station of this series' preface will briefly come back into view.

Chapter 4

The Machine

The last chapter ended with a promise of multiplication, and the cleanest way to keep it is with a picture the physicist may already own. In the great textbook of gravitation that Misner, Thorne, and Wheeler published in 1973, a tensor is introduced as a machine — a device with input slots. Into some slots you insert arrows; into others, read-outs; the machine returns a single number, and it is linear in every slot separately: double one input and the output doubles, sum two inputs in a slot and the outputs sum. That is the entire definition, and this chapter asks the reader to take it literally, because everything else follows from the slots. A covector was already such a machine, with one arrow-slot. A tensor is the same idea grown up: a machine of type (k, l) takes l arrows and k read-outs, and answers, multilinearly, with a number.

The bookkeeping comes for free, and the reader should notice that nothing new is decreed. Feed the machine basis elements of a chart and record the answers: those are the components, one number per choice of inputs, the many-indexed arrays the physicist manipulates daily. Change observers, and each slot's inputs transform by the law the last chapter derived for their species — arrows by the Jacobian, read-outs by its inverse — so the machine's components transform slot by slot, one factor of the dictionary per index. The general tensor transformation law, the one that fills a blackboard when written in full, is not a new law at all; it is Chapter 3's two laws, composed. The calculus of such component-systems is again Ricci and Levi-Civita's absolute differential calculus of 1901; the machine picture and the component picture are two presentations of one object, and the book will use whichever is honest and shorter.

The oldest tensor in physics was not proposed by a geometer; it was found inside a loaded body. Cauchy, in 1822, analyzing the internal forces of continuous matter, established the fact that deserves restating in the machine's language: cut a small surface element anywhere inside a stressed body, and the traction across it — the force per area the material on one side exerts on the other — depends *linearly* on the element's orientation. Insert the surface's normal direction into a slot; receive the traction. That machine is the stress tensor, the object from which the whole subject took its name — *tensor*, from tension — and it is the reason the word belongs to physics by birthright rather than by loan.

A debt acknowledged in passing, not paid. The stress tensor is the native object of the fluid description — the fourth station of this book's preface, the sounding promised in the founding text of the apparatus and not yet taken. It appears here as the historical headwater of the tensor concept, and this chapter spends nothing from that owed account.

A tensor field is then a tensor at every point — a machine stationed at each event of the manifold, its components varying smoothly in every chart — and with that phrase the reader already holds the form in which physics states its laws: the fields of the trade are tensor fields, stresses and fluxes and, chapters from now, curvatures.

Among the machines, one family earns a chapter-within-a-chapter: the machines that care about orientation. Ask a machine with several arrow-slots to change sign whenever two of its arrow inputs are exchanged, and what it measures is no longer a list of directions but an oriented patch — the parallelogram two arrows span, the cell three arrows span, sign flipping when the orientation flips. These alternating machines are the differential forms, and their calculus — the exterior calculus — was built by Élie Cartan beginning in 1899. Two facts about it carry this book's longest theme another step. First, forms are the natural integrands: what one integrates over a curve, a surface, a region, is precisely a form of matching degree, because integration is nothing but summing a machine's readings over the oriented cells that tile the domain. Second — and the reader should hear the theme — Cartan's derivative of a form, the exterior derivative, is defined with no metric and no further apparatus at all: from the treaty's smooth structure

alone, each k -form has a $(k+1)$ -form derivative, measuring how the machine's readings on a cell's boundary fail to cancel. No ruler was admitted, and a full differentiation arrived anyway. The inventory of what stands without lengths now includes an integral and a derivative.

Those two constructions meet in the one theorem this chapter states in full, because the reader has known its special cases since their second year: the integral of a form's exterior derivative over a region equals the integral of the form over the region's boundary. The boundary reads the interior. Every classical integral identity is this sentence in a particular costume, and the costumes have a history worth telling with the credit put right, since this series has a habit about that. The plane case is Green's, published in 1828 in a privately printed essay. The divergence case belongs to Gauss and to Ostrogradsky, who gave the general form in 1831. The curl case was communicated by Kelvin in a letter to Stokes in 1850, and Stokes set it as a question in the Smith's Prize examination of 1854 — and the trade, reading the exam rather than the letter, attached the examiner's name to the examinee's theorem, where it has stuck ever since. The general statement, uniting all of them for forms of every degree, belongs to the exterior calculus and is Cartan's. This book calls the general theorem Stokes' theorem, as the trade does, and records here whose it is.

Take the inventory once more, because it has grown quietly and the reader should audit it. Machines at a point; fields of machines; among them the oriented family, with an integral calculus and a derivative of their own; and one master identity binding derivative to boundary. Every construction cited, none original, and — the standing astonishment — still no metric: nothing yet measures a length, and none of the above required one.

What the machines cannot yet do is talk to their neighbors. A tensor at this point and a tensor at the next point live in different rooms — different tangent spaces — and nothing constructed so far compares them: subtraction of neighbors, the first gesture of differentiation for arrows and general machines, is not yet defined. The missing structure is a rule for carrying machines from room to room, it is genuinely extra — no treaty clause supplies it — and choosing it honestly is the next chapter's whole business.

Chapter 5

The Comparison

Two laboratories sit in neighboring villages, and each holds an arrow — a measured velocity, say, each valid at its own bench. A question that sounds innocent ends the innocence of this book’s construction: are the two labs holding *the same* arrow? On the flat paper of a schoolroom the question never even registers as a question — slide one arrow over to the other, keeping it parallel to itself, and compare; everyone has done it; nobody has ever been asked to justify it. But the sliding is the justification, and on ground that is not flat it fails to be innocent: carry an arrow along the surface of the Earth from the equator to the pole by one route, and by another route, keeping it as parallel as the ground permits at every stage, and the two deliveries arrive pointing in different directions. The schoolroom’s free slide was a hidden privilege of flatness. On a general manifold, arrows at different points live in different tangent spaces — different rooms, as the last chapter said — and there is no built-in fact of the matter about which arrow here is “the same” as which arrow there. If the two labs want to compare, a rule for carrying must be *chosen*. This chapter is about that rule: what choosing one means, what it costs, and who wrote its grammar.

State the honest position first, because the entire chapter rests on it: nothing constructed so far supplies the rule. The treaty of Chapter 2 lets observers translate descriptions of one point; it says nothing about relating one point to the next. The machines of Chapter 4 operate at their stations. A rule for carrying arrows along curves — the trade calls it a connection, and the carrying itself parallel transport — is genuinely additional structure, laid on top of the smooth manifold by decision, exactly as the topology was a declared inventory of regions and the smooth structure a declared treaty of

observers. The subject's honesty lives in remembering which structures were declared and when.

What must a carrying rule provide? Along any curve, a prescription taking an arrow at the starting point to an arrow at each later point, linearly — sums carry to sums, scalings to scalings — and consistently with the calculus already in hand. The working form of the demand is a derivative: given a field of arrows and a direction to probe in, the rule must produce the honest rate of change of the field — the covariant derivative — obeying linearity and the Leibniz product rule. In a chart, the naive gesture — differentiate each component function and call the results the derivative's components — fails the observers' treaty: transition maps bend, and the naive components of change do not transform as an arrow's components must. The repair is a correction term, one array of coefficients that records, chart by chart, how much the rule's notion of “held parallel” disagrees with the chart's notion of “components held constant.” Those coefficients are the Christoffel symbols, published by Elwin Bruno Christoffel in 1869 in his study of how differential expressions transform — and an honesty note belongs to them immediately, because the trade's notation invites the error: the symbols are not the components of any tensor. They are the coordinates of the *rule*, not readings of a machine; under a change of chart they transform with an extra, inhomogeneous term — the signature of a dictionary, not of an object. Confusing the rule's bookkeeping with a physical machine is the classic beginner's injury, and the cure is the distinction this book has enforced since Chapter 3: machines transform by the two laws; the symbols do not, because they are not machines.

The rule's geometric soul was exhibited by Tullio Levi-Civita in 1917, and his construction deserves its picture. For a surface sitting inside ordinary space, carry an arrow along a curve by the only procedure available to an honest inhabitant: at each instant, let the arrow ride unchanged in the surrounding space, then project away the component the surface cannot hold — keep what is tangent, discard what points out of the world. The result — parallel displacement, in his phrase — is a carrying rule derived from the surface's shape alone, and it is the rule under which the equator-and-pole experiment above gives its route-dependent answers. Levi-Civita's picture did for connections what the diagrams of Chapter 4 did for tensors: it made the abstraction walkable. The general concept — a connection as a structure in its own right, on any manifold, tied to no surrounding space and no metric — was forged by Élie Cartan in the early 1920s, his 1923 memoir on affinity

connected manifolds standing at the head of that line; the axiomatic form in which the covariant derivative is now taught descends from Koszul, and lives in the same standard references this series has cited since Chapter 1.

One forward clause, so the reader knows a debt when they see one paid later: among all carrying rules, once a metric arrives — chapters from now — exactly one will be singled out as the unique rule that is torsion-free and preserves the ruler's lengths under carrying; the trade calls that the fundamental theorem of the subject, and this book will collect it where the metric is bought, not before. Until then the connection stays what this chapter built: chosen, not given; extra, not implied; honest about being a decision.

And now the question that hands the next chapter its subject. A rule for carrying invites an experiment no reader will be able to resist: choose a closed route — out from the bench, around, and home again — and carry an arrow all the way around it, faithfully, by the rule. The arrow comes home. Does it come home *as itself*? On the schoolroom's flat paper, always. On the Earth of the equator-and-pole experiment, the reader already knows the answer is no — the arrow returns turned. The amount of the turning, per loop, is not an accident of the route's clumsiness; it is a measurement, it is linear in the loop's little spans, and the machine that answers it — the machine that eats a small closed loop and reports what carrying around it does — is the object this entire construction has been walking toward. The next chapter builds it.

Chapter 6

The Loop That Fails to Close

Now run the experiment the last chapter could not resist proposing. The pedagogy is the great gravitation textbook's own — Misner, Thorne, and Wheeler teach curvature by exactly this picture — and it goes like a bench procedure. Mark a small closed circuit on the manifold: out along one span, across along a second, back along the first, home along the second, a little parallelogram of edges. Take an arrow, and carry it around the circuit by the chosen rule of Chapter 5, faithfully, never letting it turn relative to the rule's own notion of parallel. The arrow comes home. Compare it with the arrow that stayed.

On the schoolroom's paper the two agree, always, and that agreement is what the word flat will now be earned to mean. On the Earth of the last chapter's experiment they do not agree: the traveler returns rotated, and — this is the fact that turns a curiosity into an instrument — the disagreement is lawful. Shrink the circuit and the turning shrinks in proportion to the little parallelogram's area; double the carried arrow and the difference doubles; swap the circuit's two spans — traverse it the other way around — and the turning reverses sign. Linear in the arrow carried, linear in each span, anti-symmetric under their exchange: the reader has spent two chapters learning what answers to that description. It is a machine. There is a tensor whose slots take the carried arrow and the loop's two spans, and whose answer is the change the loop inflicts; it is called the Riemann curvature tensor, and it is the object this construction has been walking toward since the first chart was laid down.

The trade reaches the same machine by a second route, and the two routes teach different things, so the book states both. Instead of one loop, take the

two probes separately: differentiate a field of arrows east then north, by the honest derivative of Chapter 5, and again north then east, and subtract. On flat paper the order of probing never matters; on curved ground it does, and the failure of the two second derivatives to commute is, component for component, the same tensor the loop measures. That is the form the absolute differential calculus of Ricci and Levi-Civita computes with — curvature as the commutator of covariant derivatives — and it makes plain a fact the loop picture hides: no metric was consulted. The carrying rule alone — the declared structure of Chapter 5, no ruler anywhere — suffices to define curvature, measure it, and compute with it. The longest-running theme of this book reaches its summit here: the celebrated tensor of curved geometry stands complete on a manifold that still does not know what a length is. And the word flat now has its earned meaning: a carrying rule is flat when every loop returns every arrow as itself — when the machine reads zero on all inputs everywhere.

The history is Riemann's, and it is the second half of a story this book already began. The lecture of 1854 posed the question in nearly this form: if space is an extended magnitude charted region by region, what measures its deviation from flatness? The apparatus that answers — the four-index expression the trade now writes — appears in his essay of 1861 for the Paris Academy, on an applied problem of heat conduction, where the geometry enters as the tool; and like the lecture, the essay was published only after his death, in the collected works his friends assembled. Riemann held the whole subject — the charts, the question, and the answering machine — and lived to publish almost none of it. The reader who wants the machine's finer anatomy — its symmetries, which cut its many components down to few, and its standard contractions, the smaller readings that carry Ricci's name and, contracted once more, yield a single scalar — will find them in the standing references; this book takes from the anatomy only what later chapters spend.

An illustration, not an identity — and the last device appearance in this book. The apparatus documented in Volume 1, built to describe its own act of describing and to close that circuit on itself, returns from the closed circuit changed: a residue survives the loop, reproducibly, refusing to vanish however the descent is refined. That is the same shape the reader has just built — a carrying around a closed loop that fails to return its object as itself — and it is offered here as an illustration of the shape and as nothing else. The device does not thereby compute a curvature, and this book does not claim it does; whether that reading could ever be earned is a question that stays on the far side of a fence this series has not crossed — a face still owed, like the fluid's. Illustration, not identity.

Take the chapter's inventory, because it completes an arc. A machine that measures the failure of loops to close; a second route to it through the failure of probes to commute; a definition of flatness with honest content; and all of it — the reader should say this aloud, because few textbooks pause on it — without a single length. Nothing so far prefers one scale to another, no angle has been measured, no distance compared, and yet charts, arrows, machines, carrying, and curvature all stand. The construction has run out of things it can build without a ruler exactly at the moment the reader can feel what a ruler would be for.

So the next chapter, at last, goes shopping. It buys the metric — the smooth choice of an inner product at every point — and the reader will watch what the purchase settles all at once: lengths and angles, the marriage of arrows to read-outs promised in Chapter 3, and, by the fundamental theorem promised in Chapter 5, the selection of exactly one carrying rule from among all the candidates. The ruler arrives last because everything before it never needed one; what it buys is the subject of the next chapter, and the price is paid in the open.

Chapter 7

The Ruler

Audit the shop before the purchase, because the purchase is the point. Charts and their treaty; arrows and the read-outs that measure them; machines of every slot-count; a rule for carrying, chosen in the open; and a machine that reads the failure of loops to close. Every station of that inventory was built without measuring a single length — the reader was asked, chapter by chapter, to notice the absence — and the construction has now genuinely run out of things it can build without one. So this chapter, at last, goes shopping, and the book keeps its standing promise: the price of everything is paid in the open.

What is a ruler, in the language this book now owns? A machine. At each point, install a machine with two arrow-slots — symmetric, so the slots do not care about their order, and positive, so that no arrow but the zero arrow reads zero against itself — and let the installation vary smoothly from point to point in every chart. That field of two-slot machines is the metric tensor. Feed it the same arrow twice and the answer is declared to be the arrow's squared length; feed it two different arrows and the answer carries what the trade distills into the angle between them. The proposal is Riemann's, from the 1854 lecture this book keeps returning to: geometry enters not as a global grid but as a length-rule given infinitesimally — his line element — point by point, chart by chart, with which geometry a given space actually wears left, in his own insistence, to be determined by experience. Everything in this chapter descends from that one installation, and the standing references for the descent are the subject's standard texts — do Carmo's *Riemannian Geometry*, Lee's *Riemannian Manifolds* — which this book cites once and then declines to re-derive.

Now watch what the purchase settles, because three debts fall due at once, and each was flagged, chapters ago, as a debt.

First, lengths themselves. A curve gains a length: integrate, along it, the length the metric assigns its velocity — the speed — and distances follow as the infimum of curve-lengths between two points. The paths that locally spend that length most frugally are the geodesics, the subject's straight lines, and with them the reader's oldest geometric vocabulary — how long, how far, how straight — is finally licensed on curved ground.

Second, the marriage promised in Chapter 3. An arrow and a read-out were distinct species, their conversion flagged as a purchase not yet payable. Here is the payment. Take the metric's two-slot machine and feed an arrow into one slot only, leaving the other open: what remains is a machine with one open arrow-slot — which is to say, a read-out. The metric pairs each arrow with its own covector, one-to-one, and the pairing is geometry, not notation: it depends on the installed ruler and changes if the ruler changes. The physicist's daily act of lowering an index is exactly this feeding-of-one-slot, the raising the inverse machine's, and the license the reader has used since their second year is hereby dated and priced: legitimate wherever a metric stands installed, bought at the cost of installing it, and not one chapter earlier.

Third, the selection promised in Chapter 5. Among all the carrying rules a manifold admits — and Chapter 5 was honest that nothing so far preferred one — the ruler singles out exactly one: the unique rule that is torsion-free and that preserves the ruler itself under carrying, so that lengths and angles ride unchanged along any transport. That is the fundamental theorem of the subject, its proof in the standard texts just cited, and the selected rule bears Levi-Civita's name — the same Levi-Civita whose 1917 picture made carrying walkable — closing a loop of names this book finds pleasing but spends nothing on. From here the machinery composes: the metric selects its connection, the connection feeds Chapter 6's loop-machine, and the curvature that emerges is the curvature *of the ruler's own geometry* — Riemannian geometry proper, the discipline the 1854 lecture called for.

And a fourth settlement, smaller but owed to the very first chapter: the ruler refines the neighborhoods. Around each point the metric draws the balls of every radius — all points within distance epsilon — and these balls are neighborhoods in exactly Chapter 1's sense, a basis for the space's regions of validity. The reader's epsilon-and-ball calculus is thereby recovered whole. But the order of construction must be kept honest, because it is the moral

of the entire book: the topology was declared first, and the metric's balls *describe* the same neighborhoods more finely — they do not create the space, they instrument it. The limits of Chapter 1 needed no norm; the norm, arriving six chapters later, agrees with them.

One paragraph belongs to a sibling ruler, because the next chapter cannot be read without it. Relax the demand that the two-slot machine be positive, and allow a signature in which one direction reads opposite in sign to the rest. The machine then grades directions into kinds — those with positive reading, those with negative, and the boundary family reading exactly zero — and geometry acquires a causal grammar: the timelike, the spacelike, and the null. That modification, put before physics in Minkowski's lecture of 1908, is the geometry in which relativity writes spacetime, and every construction of this book — charts, machines, the selected connection, the loop-machine — carries over to it unchanged in form. This book needs the fact and not its development; one paragraph, as promised, and no more.

Stand back once, before the final chapters, and say what the ruler did and did not do — because the shape of this chapter is the thesis of the volume. The ruler created nothing. Charts, arrows, read-outs, machines, carrying, curvature: all of it stood before a single length was measurable, and the reader watched it stand. What the ruler did was select and measure: it married each arrow to a read-out, elected one carrying rule from the candidates, drew the balls, and put numbers to lengths, angles, and straightness. The ruler is not the ground; it is the final instrument installed on ground that stood without it. A reader who carries only that one sentence out of this book carries the construction.

The language is now complete. Neighborhoods for validity, charts for observers, arrows and read-outs for states and instruments, machines for the laws, a carrying rule for comparison, a loop-machine for curvature, and a ruler for measure — every piece cited, every price public. What remains is to read: the two great descriptions of the century, each written in this language and each favoring a different one of Chapter 3's two species, and the boundary between them where the founding text of this series located its deepest promise. That reading is the next chapter, and it ends at a door.

Chapter 8

The Two Faces at the Boundary

The language is assembled, every piece cited and every price public, and this chapter does what the whole construction was for: it reads. Two descriptions dominated the physics of the twentieth century. Both are written in exactly the language of the last seven chapters, and — the fact this book has been walking toward — each of them leans on a different one of Chapter 3’s two species. One is a covariant-face description; the other is contravariant-face. Neither face was invented for its theory; the reader watched both fall out of the chain rule, cited, before either theory entered the room.

Read the geometric description first. In November 1915 Einstein published the field equations of gravitation, and in this book’s language the sentence is short. Take the ruler — in its Lorentzian signature, the one paragraph of Chapter 7 — and let it select its carrying rule by the fundamental theorem; feed that rule to Chapter 6’s loop-machine; contract the resulting curvature to its Ricci reading and scalar, the smaller readings Chapter 6 named. That assembly — curvature, read down to two slots — stands on the equation’s left. On its right stands a two-slot machine holding the content of matter and field — the stress-energy, the direct heir of the machine Cauchy found inside loaded bodies in 1822. The equation says the two machines agree, slot for slot, at every event: geometry’s curvature reads matter’s content. And the description’s whole grammar runs on the lowered face: the metric, the Ricci reading, the stress-energy — two covector slots each, machines awaiting arrows, the covariant species of Chapter 3 doing all the load-bearing work.

Now the quantum description. In 1930 Dirac published the book that gave the theory its grammar, and the grammar is Chapter 3's other face. A state of a system is a ket — an arrow, not in the spacetime manifold but in a state-space of its own, linear from the outset; an instrument's question is a bra — a read-out, an element of the dual; and every prediction the theory makes is the elementary act this book met in Chapter 3, instrument-meets-state, the bracket that gives the formalism its name. The description's working face is the raised one: the state side, the arrow species, contravariant under its changes of basis. One honesty note belongs here, and Chapter 3 already paid for it: what the two descriptions share is the *grammar* — the two species and their pairing — not the stage. The quantum arena is not the spacetime manifold, and this book does not blur the two; it observes only that both theories speak the constructed language, each favoring one face.

Between the two faces stands one more cited result, and it is the closest thing the mathematics owns to a stitch. In 1927 — before Dirac's book — Erwin Madelung showed that the Schrödinger equation for the electron can be rewritten, exactly, as a pair of fluid equations: a continuity equation for a density, and a momentum equation of Euler's form carrying one additional term the trade calls the quantum potential. Not an analogy; a change of variables. The electron's quantum description genuinely has a hydrodynamic form, and the reader of this series' preface will recognize what that means: the fourth station — the fluid description, the sounding not yet taken — has, in Madelung, its nearest earned neighbor. What separates Madelung's fluid from the fluid of the founding promise is, among other things, viscosity: the dissipative structure of the Navier–Stokes equations is not in the 1927 result, and this book, which has paid every debt in the open, marks this one still owed.

The founding program, stated as what it is. The founding text of the apparatus this series documents — the same opening pages that owed the fluid its debt — sets the series its deepest assignment in one sentence: that the covariance of the geometric description will never settle, at the boundary, with the contravariance of the quantum one, no matter how much energy is there. And the text fences itself in the next breath, allowing the empty case: at zero energy there is nothing to disagree about. This book states that sentence as what it is — a promise the series' reader is to be shown, made by the founding text, in words — covariance, contravariance — that Chapter 3 of this book constructed from the chain rule and cited to their owners. It is not a theorem. No volume of this series has proved it, this one least of all; the two readings above show only that both descriptions speak one language while favoring opposite faces — which is what makes the boundary sayable, and nothing about what happens there. Whether the promise can be earned, and in which volume's terms, stays on the far side of the same fence that holds the fluid: owed, named, and not crossed here.

And that is the door at which this book ends. It was never this volume's work to settle a boundary; its work was to build, from cited topology and cited linear algebra, a language in which charts, arrows, machines, carrying, curvature, and measure all stand with their prices published — and then to demonstrate the language by reading the century's two great descriptions in it, one on each face. The reader now holds what the preface promised: the ground the four soundings stand on. Three of the four stations have appeared in these pages — the geometric, the quantum, and, at one remove, the electron model whose measured number this series' first volume brackets; the fourth is owed, and the door it waits behind is now, at least, a door with a frame. What remains for this volume is only its accounting: the ledger of names and dates this book has spent, chapter by chapter, which is the back matter's business and the next and final chapter.

Chapter 9

The Ledgers

This book promised, on its first page, that nothing in it would ask the reader to believe anything new, and a promise like that is audited, not applauded. The audit is this chapter. Three ledgers: the names and dates this book spent, chapter by chapter; the physical descriptions it borrowed as openings; and the honest grades of the faces it used. If any sentence of the preceding chapters cannot be traced to a row below or to the reader's own bench practice, that sentence is a defect, and the book asks to be held to exactly that standard.

The citation ledger

Preface

- Amontons, 1699 — the laws of friction; Coulomb, 1785 — the separation of static from sliding. Back: the stuck block.
- Lenard, 1902 — the photoelectric observations; Einstein, 1905 — the quantum explanation. Back: the red lamp that never works.
- Hooke, 1678; Young, 1807; Cauchy, 1822 — the linear zone, its modulus, and the stress–strain formalism. Back: the bar and its yield.
- Joukowsky, 1898 — the water-hammer surge. Back: the pipe that bangs.

- Classical surveying arithmetic, with satellite navigation as the living instance. Back: the trilateration earn of the number four.

Chapter 1 — The Local

- Hausdorff, 1914 (*Grundzüge der Mengenlehre*) — the neighborhood axioms.
- Kuratowski, 1922 — the closure axiomatization.
- Kelley, 1955 (*General Topology*) — the assembled standard.
- Fréchet, 1906 — abstract convergence before the neighborhood axioms.
- Alexandroff and Hopf, 1935 (*Topologie*) — the separation taxonomy; Kolmogorov's T0 identification therein.

Chapter 2 — The Chart

- Riemann, 1854 — the Habilitation lecture (published 1868 by Dedekind).
- Whitney, 1936 — the precise patchwork definition and the embedding in dimension $2n+1$.
- Lee (*Introduction to Smooth Manifolds*) — the modern standard.

Chapter 3 — The Direction

- Grassmann, 1844 (*Ausdehnungslehre*) — the vector space of directed magnitudes.
- Ricci and Levi-Civita, 1901 — the absolute differential calculus; the two transformation laws as a calculus.
- Dirac, 1930 (*The Principles of Quantum Mechanics*) — the bra-ket pairing as the dual space in the physicist's house.
- Lee (*Introduction to Smooth Manifolds*) — the tangent-space formalization, carried over from the previous chapter's citation.

Chapter 4 — The Machine

- Misner, Thorne, and Wheeler, 1973 (*Gravitation*) — the tensor as a machine with slots.
- Cauchy, 1822 — the stress tensor, the subject's founding physical instance.
- Cartan, 1899 — the exterior calculus; also the general boundary theorem.
- Green, 1828; Gauss and Ostrogradsky, 1831; Kelvin, 1850, and Stokes, 1854 — the classical cases of the boundary theorem, with the attribution told whole in the chapter.

Chapter 5 — The Comparison

- Christoffel, 1869 — the symbols; the transformation of differential expressions.
- Levi-Civita, 1917 — parallel displacement.
- Cartan, 1923 — the connection freed from the surrounding space.
- Koszul, via the standard texts — the axiomatic covariant derivative.

Chapter 6 — The Loop That Fails to Close

- Misner, Thorne, and Wheeler — the loop pedagogy.
- Riemann, 1854 and 1861 — the question, and the four-index apparatus of the Paris essay (both posthumous).
- Ricci and Levi-Civita, 1901 — curvature as the commutator; the contractions bearing Ricci's name.

Chapter 7 — The Ruler

- Riemann, 1854 — the line element.
- do Carmo (*Riemannian Geometry*); Lee (*Riemannian Manifolds*) — the standard descent, including the fundamental theorem.

- Levi-Civita, 1917 — the selected connection’s name and picture.
- Minkowski, 1908 — the signature of spacetime, one paragraph as promised.

Chapter 8 — The Two Faces at the Boundary

- Einstein, November 1915 — the field equations, read on the covariant face.
- Dirac, 1930 — the quantum grammar, read on the contravariant face.
- Madelung, 1927 — the hydrodynamic form of the Schrödinger equation; the owed face’s nearest earned neighbor.
- The founding text of the apparatus this series documents — cited as apparatus, once, for the program its opening pages set and fence.

The physical-descriptions ledger

The openings of these chapters borrowed the reader’s own practice, and practice has provenance too: the locality of every bench report (the working tradition of experimental physics); the observer’s coordinates (the same tradition, formalized by the geometers named above); velocity as the pointed question (mechanics since Newton); the machine with slots (Misner, Thorne, and Wheeler’s pedagogy, borrowed with its name); stress as force-per-area across an oriented cut (Cauchy); the carried arrow and the loop experiment (the differential geometers’ standard picture, told in the gravitation textbook’s form); the ruler and the ball (the surveyor’s and the analyst’s common stock). None of these descriptions is original here, and none carries a claim beyond the cited mathematics it introduces.

The face-grade ledger

This volume used, and here grades, exactly what it touched. Its two *inputs* were cited mathematics whole: topology (Chapter 1 — the neighborhood axioms through the T0 identification) and linear algebra at a point (Chapter 3 — spaces, duals, multilinearity). Its *subject* — the thing constructed — was

differential geometry (Chapters 2 and 4 through 7), assembled from those inputs with every step cited. Its *readings* were the two faces of Chapter 8: the geometric description on the covariant face, the quantum on the contravariant, each cited to its owner. And one face remains *owed* across the whole series: the fluid description, seeded in the founding text, nearest-neighbored by Madelung, its dissipative body undelivered — named at its two natural sites in this book and spent at neither.

One bookkeeping distinction, recorded so no reader conflates two senses of one word: the linear algebra of Chapter 3 is a cited mathematical input of this volume, and it is not the same thing as the arithmetic this series' measuring apparatus uses internally, which is exact rational arithmetic on counts and is documented in Volume 1. The two live in different ledgers — one belongs to the mathematics this book teaches, the other to the instrument this series measures with — and this row exists to keep them apart.

The closing line

The audit is the book. A volume built under the law that no original idea may appear in it succeeds exactly when its back matter is exhaustive and its chapters are traceable — when the reader can put a finger on any sentence and be pointed, by these ledgers, to a name, a date, or their own laboratory habit. The mathematics belongs to the names above. The assembly — the order of chapters, the openings, the insistence on prices paid in the open — is all this book claims, and an assembly is not an idea; it is a reading plan. The four soundings belong to the volumes around this one. The ground now holds.